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MLR-Bench: Evaluating AI Agents on Open-Ended Machine Learning Research

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: https://github.com/chchenhui/mlrbench

Abstract

Recent advancements in AI agents have demonstrated their growing potential to drive and support scientific discovery. In this work, we introduce **MLR-Bench**, a comprehensive benchmark for evaluating AI agents on open-ended machine learning research. MLR-Bench includes three key components: (1) 201 research tasks sourced from NeurIPS, ICLR, and ICML workshops covering diverse ML topics; (2) **MLR-Judge**, an automated evaluation framework combining LLM-based reviewers with carefully designed review rubrics to assess research quality; and (3) **MLR-Agent**, a modular agent scaffold capable of completing research tasks through four stages: idea generation, proposal formulation, experimentation, and paper writing. Our framework supports both *stepwise* assessment across these distinct research stages, and *end-to-end* evaluation of the final research paper. We then use MLR-Bench to evaluate six frontier LLMs and an advanced coding agent, finding that while LLMs are effective at generating coherent ideas and well-structured papers, current coding agents frequently (e.g., in 80% of the cases) produce fabricated or invalidated experimental results—posing a major barrier to scientific reliability. We validate MLR-Judge through human evaluation, showing high agreement with expert reviewers, supporting its potential as a scalable tool for research evaluation. We open-source MLR-Bench to help the community benchmark, diagnose, and improve AI research agents toward trustworthy and transparent scientific discovery.

1 Introduction

A hallmark of human intelligence is the ability to generate new knowledge through research and scientific discovery. From basic research to applied technological development, scientific progress has shaped human civilization. Reproducing this ability—to autonomously generate, test, and validate new knowledge—has long been viewed as one of the grand challenges for artificial intelligence.

Recent advances in large language models (LLMs) bring us closer to this vision. LLM-powered agents have demonstrated impressive capabilities in generating research ideas [16, 3, 28, 17], conducting experiments and analyses [11, 34, 4, 40, 13, 41, 15], drafting scientific articles [25, 35, 38, 1] and automated review [36, 14, 32, 6, 42]. These advances move us beyond automating isolated tasks, opening up the possibility of automating the entire scientific process, from idea conception to experimentation to dissemination.

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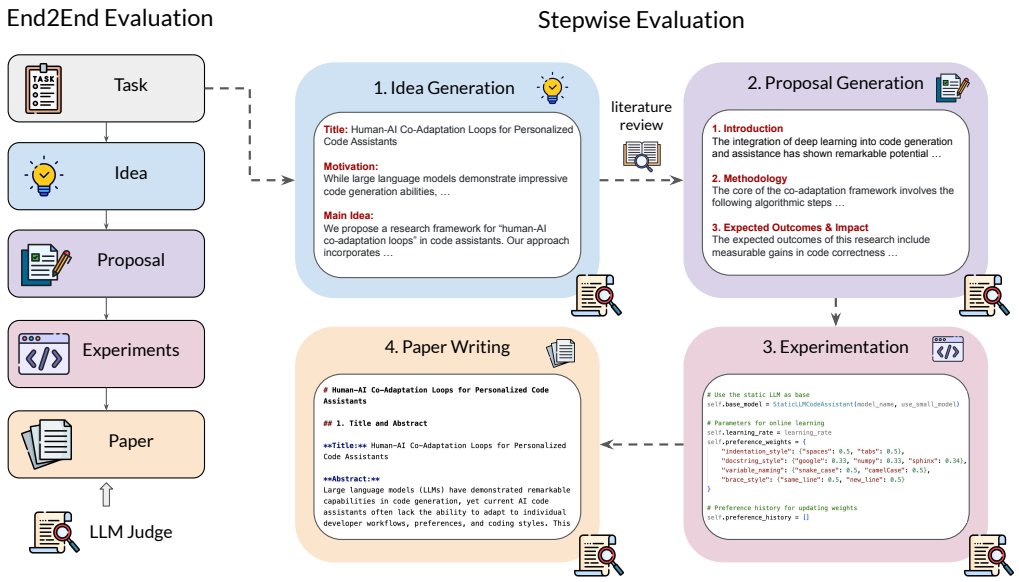

Figure 1: An overview of the framework of MLR-Bench, consisting of both an end-to-end evaluation (left) and a stepwise evaluation (right), each of which uses LLM judges to automatically assess performance over 201 tasks. For end-to-end evaluation, we use the same model as backbone in idea generation, proposal generation and paper writing. For stepwise evaluation, various models are tested and compared within each step.

However, realizing this vision raises an important question: *how do we rigorously evaluate the quality of research produced by AI agents?* While recent works have shown promising results [29, 27, 19, 26, 39], the community still lacks a comprehensive benchmark to systematically assess AI agents' ability to conduct open-ended scientific research, making progress toward autonomous scientific discovery difficult to measure and compare fairly between agents. Moreover, there is also a need for empirical analysis to identify key failure modes—such as hallucinated results, lack of novelty, or methodological flaws — to quantify current limitations and inform future progress in research agents.

Toward this vision, we introduce **MLR-Bench**, a comprehensive benchmark designed for evaluating AI agents on *open-ended machine learning research tasks*. Specifically, we seek to answer the following research questions:

- **RQ1:** *How well can AI agents conduct open-ended machine learning research?* (§3)
- **RQ2**: *How effectively can an LLM judge evaluate research, as measured by its agreement with human reviewers?* (§4)
- **RQ3:** *What are the key factors that affect the quality of AI-generated research? (§5)*

To systematically address these questions, we decompose the research process into four successive stages: (1) *idea generation*, (2) *research proposal formulation*, (3) *experimental execution and analysis*, and (4) *paper writing*. This decomposition allows us to assess both stepwise progress and the overall end-to-end research capability of AI agents.

MLR-Bench consists of three key components: (1) a collection of 201 real-world research tasks sourced from NeurIPS, ICLR, and ICML workshops over the past three years, covering a wide range of machine learning domains, including LLMs, trustworthy AI, ML systems, AI for science; (2) **MLR-Judge**, an automated evaluation pipeline consisting of LLM-based judge and structured review rubrics; and (3) **MLR-Agent**, a modular research agent scaffold capable of automatically completing these tasks by following the four defined research stages.

To explore RQ1, we use MLR-Bench to evaluate six state-of-the-art models—including the most recent `o4-mini` [24], `Gemini-2.5-pro-preview` [8], and `Qwen3-235B-A22B` [31]—as well as the advanced coding agents, `Claude Code` [2] and `Codex` [23]. Our results show that while these models perform well in generating ideas and writing papers, they often fail to deliver innovative

or scientifically reliable research, mostly due to coding agents producing invalidated experimental results.

To validate the reliability of MLR-Judge (addressing RQ2), we recruit 10 machine learning experts with top-tier conference review experience to independently review the AI-generated research outputs. We find that the level of agreement between an LLM judge and a human judge, when assessing the same output, is very close to the level of agreement between two human judges, demonstrating its effectiveness as a reliable evaluation tool. Finally, to address RQ3, we analyze the justifications provided by both MLR-Judge and human reviewers, identifying a recurring failure mode where agents tend to generate fabricated or unverified results after execution failures—revealing a gap between fluent output generation and true scientific rigor.

Our contributions include:

- **MLR-Bench**: To our knowledge, the most comprehensive evaluation benchmark for AI research agents to date, featuring 201 open-ended ML research tasks, a human-aligned **MLR-Judge** for automated research quality assessment, and a modular **MLR-Agent** supporting both stepwise and end-to-end research execution.
- **Comprehensive Evaluation and Insights**: We apply MLR-Bench to evaluate 6 frontier models, providing comparative insights into their strengths and limitations in scientific discovery. Also, we compare MLR-Agent with AI Scientist V2 on MLR-Bench, and find existing research agents exhibit critical failure modes, including the widespread issue of fabricated experimental results and hallucinated methodology, pointing to key challenges for future research.

2 MLR-Bench

In this section, we describe the overall flow of MLR-Bench. Fig. 1 presents an overview of the framework of MLR-Bench, which consists of both an end-to-end evaluation pipeline to evaluate AI agents' ability to complete end-to-end research, and a stepwise evaluation pipeline, which separately evaluates AI agents' abilities on (1) Idea Generation, (2) Proposal Generation, (3) Experimentation, and (4) Paper Writing.

Tasks. To measure the performance of AI agents across various machine learning topics, we collect 201 tasks from ICLR, ICML, and NeurIPS workshops over the past three years. These tasks are categorized into 9 core ML topics: LLMs and Vision Language Models (VLMs), AI for Science, ML Theory, Trustworthy AI, Computer Vision, ML Systems, Multimodality, Reinforcement Learning (RL), and other emerging topics. Fig. 2 illustrates the distribution of our tasks across these ML topics. To curate our dataset, we first review all workshops from ICLR, ICML, and NeurIPS conferences over the past three years. We then filter out duplicate workshops and select those that maintain complete information and target a general audience. Finally, we extract workshop overviews and topics to formulate our tasks.

End-to-End Evaluation. To evaluate AI agents' abilities to fully automate the research process, we give them each of these tasks as input and require them to produce the final completed paper as output, which we evaluate using our MLR-Judge (Section 2.1). To facilitate software experimentation, agents are provided with an environment that includes file system access, a Python runtime for executing scripts, and internet connectivity.

Stepwise Evaluation. To assess AI agents' research capabilities in a more fine-grained manner, we evaluate them over 4 steps, using MLR-Judge to assess each step: (1) *Idea Generation:* we provide the agent with a research task and require it to generate a research idea. (2) *Proposal Generation:* we provide the agent with a research task and idea, and require it to generate a detailed research proposal. (3) *Experimentation:* the agent is given access to an environment for running experiments (similar to in the end-to-end case), and iteratively writes code and runs experiments. (4) *Paper Writing:* agents are given a full set of experimental outputs (report, figures, and a log of commands run during experimentation), and required to produce the final paper. This task involves multimodal input and output, so we test multimodal agents in this stage.

Each step of the stepwise evaluation requires a set of **input data** for the agents. For example, in step (1), the *Idea Generation* agents require research tasks as input data; we use the same 201 tasks used in